

65TH ERSACONGRESS

Decoding Place Functionality

*Identifying Emerging Third Places
and the Drivers of Remote Work Visitation
using GPS signals and Google Places Reviews*

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How WFH changes cities

Three channels through which remote-work behavior reshapes urban form and public services.

0 1



Commuting behavior

Less load on public transit

Residents stay home more days per week — reducing peak-hour congestion and freeing capacity on the public-transit network.

0 2



Residential choice

Re-sorting of where people live

No premium for proximity to the CBD: residents move toward suburbia and amenity-rich periphery, redrawing the rental price gradient.

0 3



Local development

Growth of neighborhood services

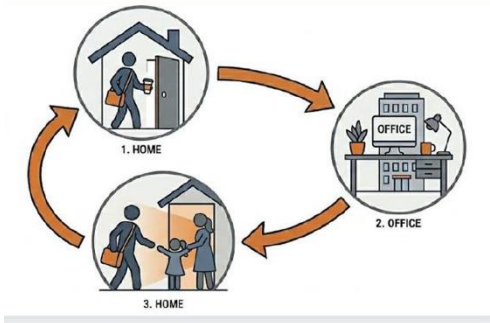
Demand for local cafés, shops, parks and community space rises — triggering new openings, mixed-use zoning and infrastructure upgrades.

From "home-work-home" to "home-third place-home"

COVID-19 reshaped the urban triad. Hybrid schedules brought ~30% of US knowledge work home — and turned cafés, libraries and community centers into office substitutes.

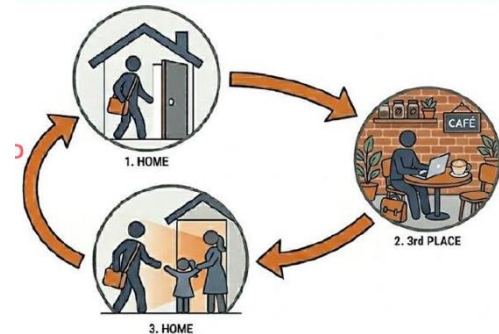
PRE-COVID TRIAD:

commute



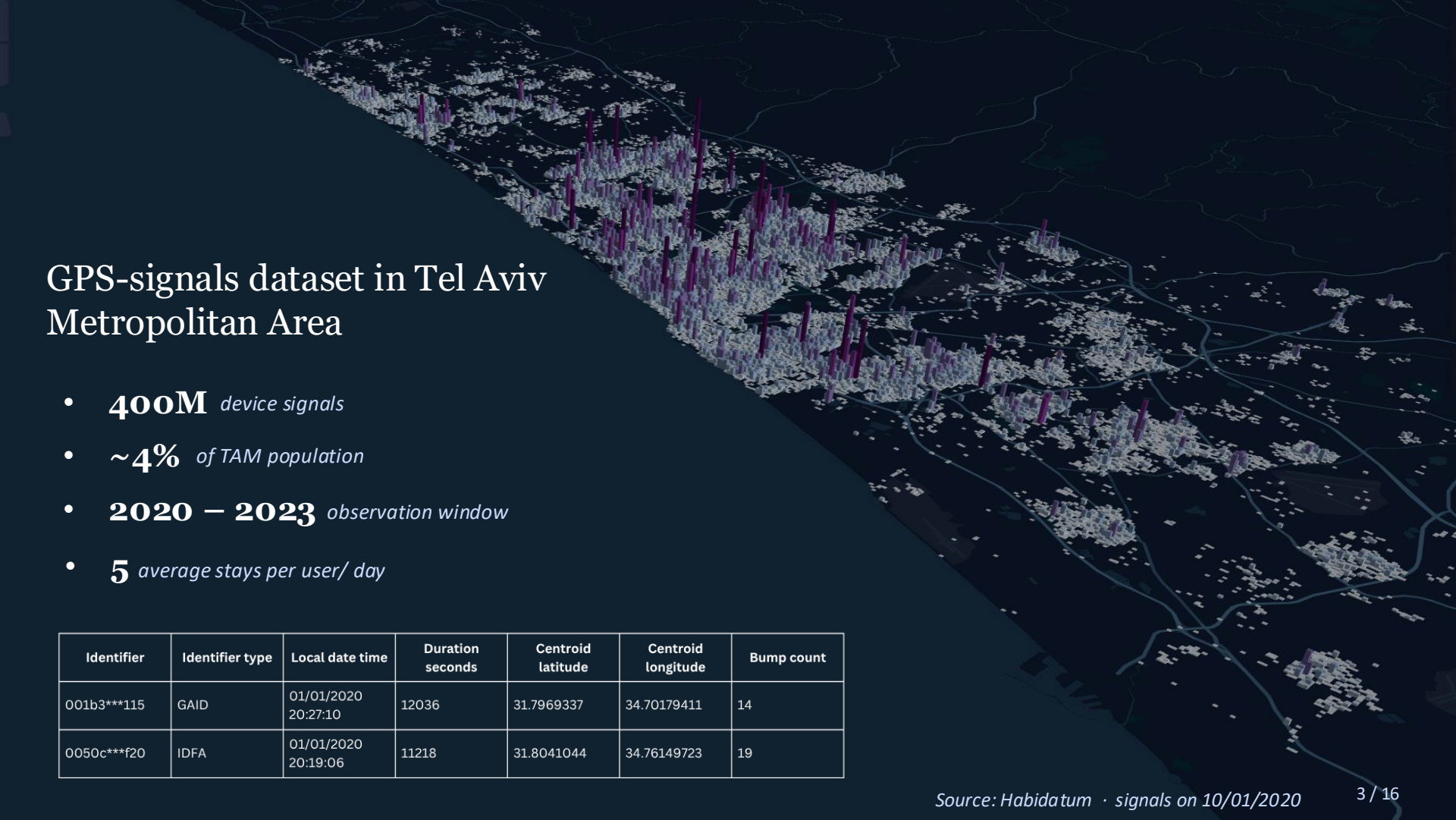
POST-COVID PATTERN:

local activity?



RESEARCH QUESTION

How do 'new third places' change residential behavior after COVID-19?

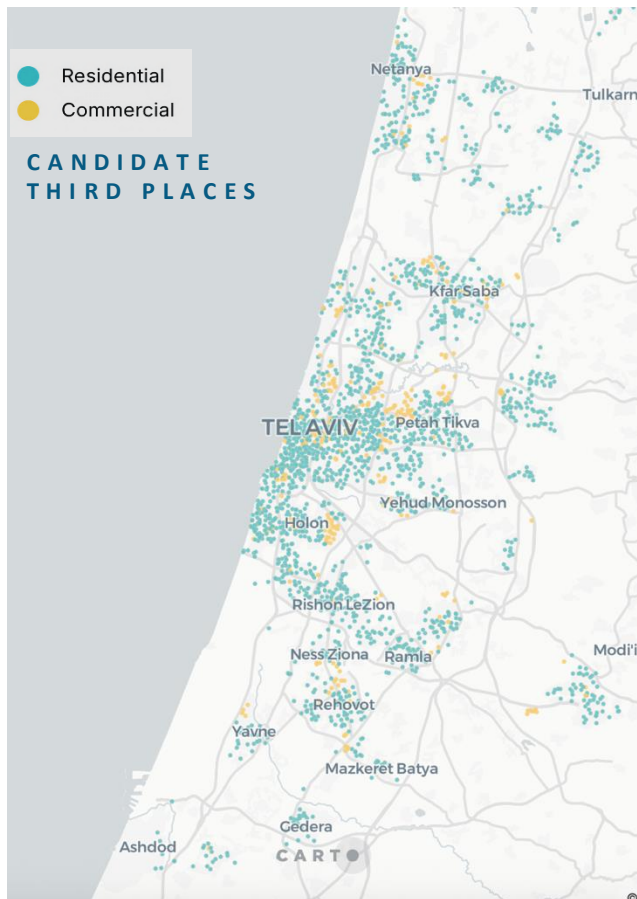
An aerial map of the Tel Aviv Metropolitan Area, showing a dense urban landscape. Overlaid on the map are numerous purple, vertical, spike-like markers of varying heights, representing GPS signals. The spikes are most concentrated in the central urban core and along major transportation corridors. The background is a dark, textured map with light-colored lines indicating roads and boundaries.

GPS-signals dataset in Tel Aviv Metropolitan Area

- **400M** *device signals*
- **~4%** *of TAM population*
- **2020 – 2023** *observation window*
- **5** *average stays per user/ day*

Identifier	Identifier type	Local date time	Duration seconds	Centroid latitude	Centroid longitude	Bump count
001b3***115	GAID	01/01/2020 20:27:10	12036	31.7969337	34.70179411	14
0050c***f20	IDFA	01/01/2020 20:19:06	11218	31.8041044	34.76149723	19

Points of interest · Google Places + OSM + Israel Census



CANDIDATE THIRD PLACES

Cafés, libraries, coffee shops, bakeries, museums, community & cultural centers, bookstores

Google Places API

3,524

WORK-ZONE & TRANSPORT HUBS

Business centers, corporate offices, industrial areas, universities, schools

Google Places + OSM + Census

1,378

OTHER AMENITIES

Hospitals, Parks, Retail, Entertainment, Outdoor attractions, Lifestyle services

OpenStreetMap

~34K

STATISTICAL AREA CONTEXT

- Population
- income
- Religiosity
- Occupation WFH potential

Israel Census 2022 + Soc-Economic Survey 2019

602

Three-steps methodological framework

The study links a machine-learning identification step to a behavioral and an econometric analysis of impact.

PART 1

1

Identification

Detect the third places

Google Reviews + a Positive-Unlabeled
classifier

PART 2

2

Behavior analysis

Compare visits to Third places vs rest POIs

Descriptive Statistics

PART 3

3

Impact

*Estimate effects of third places on local
residents POIs visits (hours)*

WLS Regressions

Part I· Identifying third places : Ground truth and features from Google reviews

Review-derived features

-Number of reviews -Average Rating -Price range -Serve coffee -Restroom

Direct evidence

✓ INCLUDE

'laptop'
'place to/for work'
'remote work'
'business meeting'
'computer'
'get the work done'

✗ EXCLUDE

'event venue'
'no Wi-Fi'
'no laptop'
'on the go'
'repair'

Flag 'Suitable ambience'

✓ INCLUDE

'quiet'
'relaxing'
'Comfortable'
'calm'
'meeting friends'
'sit with'
'wifi'
'charger'

✗ EXCLUDE

'not quiet'
'rush'
'loud'
'not comfortable'
'not relaxing'
'not calm'

100

places with explicit 'work-from-here' evidence

Classification model



PU learning (Elkan – Noto)

Elastic-net logistic regression with PU adjustment.

Trusted positives only — 'unlabeled' rest

7 features × 3,524 candidates

100 iterations × 6 hyper-parameter grids

Validate: hide 30 % positives, recall@k

$$P(y=1|x) = P(s=1|x) / c$$

Ranked probability output

Detected places suitable for remote work

Part 3 · Do local Third-Places Reshape Commercial Activity Post-COVID? Models Specification

Four stepwise WLS specifications with two-way fixed effects. We compare January 2020 with Q12023

M1

Baseline Distance Decay

Post-shock shift in distance-decay

$$Y_{it} = \beta_1 \cdot (Post \times Dist) + FE_t + FE_s + \varepsilon$$

M2

Remote Work Capacity

Does IT-workers share anchor local mobility?

$$Y_{it} = \beta_1 \cdot (Post \times Dist) + \beta_2 \cdot (Post \times WFH_{pot}) + \beta_3 \cdot (Post \times Dist \times WFH_{pot}) + FE_t + FE_s + \varepsilon$$

M3

Third places impact on local activity

Localized amenity concentration

$$Y_{it} = \beta_1 \cdot (Post \times Dist) + \beta_4 \cdot (Post \times Dose) + \beta_5 \cdot (Post \times Dist \times Dose) + FE_t + FE_s + \varepsilon$$

M4

Third places vs WFH impact

WFH vs amenity, side by side

$$Y_{it} = \beta_1 \cdot (Post \times Dist) + \beta_2 \cdot (Post \times WFH_{pot}) + \beta_3 \cdot (Post \times Dose) + \beta_4 \cdot (Post \times Dist \times Dose) + \beta_5 \cdot (Post \times Dist \times WFH_{pot}) + FE_t + FE_s + \varepsilon$$

Part 3 · Do local Third-Places Reshape Commercial Activity Post-COVID? Model targets

We are testing several hypothesis by comparing Q12020 (Jan, Feb) with Q12023

Do local* Third places impact...

substitution of “global” POIs for local ones?

Local POI hours / all POI hours

the amenities travel radius?

Radius of Gyration*,POIs

the preference for local TP candidates*?

Local candidate hours / all candidate hours

the shift of local time from home to amenities?

Local POI hours / all POI hours

the overall affinity for out-of-home activities?

POI hours / all hours

* Candidates – Google Places in categories commonly used for remote work

* Local area – 1km buffer around each mobile user home locations

* Radius of Gyration- the RMS distance from the center of users activity

Part 1. Identification results

538 places identified

15% of all candidates

Tel Aviv
coffee_shop **34%**

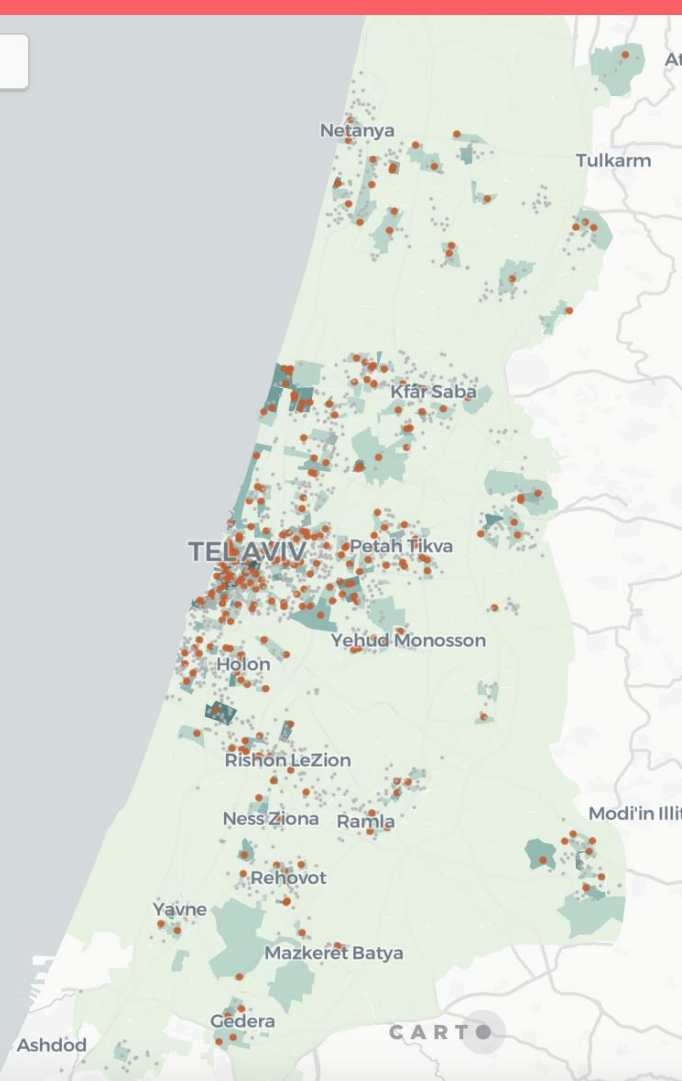
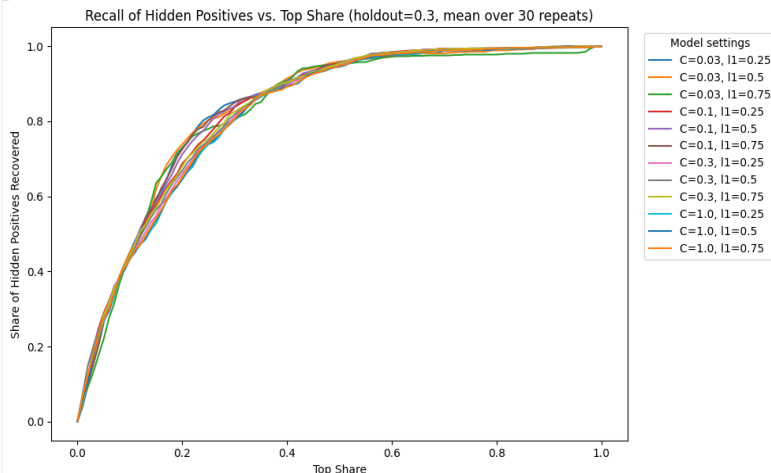
77% recall at top 20

Satellite cities
Libraries **38%**

BY POI TYPE

POI	N	RECALL
Coffee shop	171	71 %
Cafe	128	50 %
Library	47	100 %
Cultural center	62	100 %
Bakery	63	33 %
Community center	27	—
Book store	20	100 %

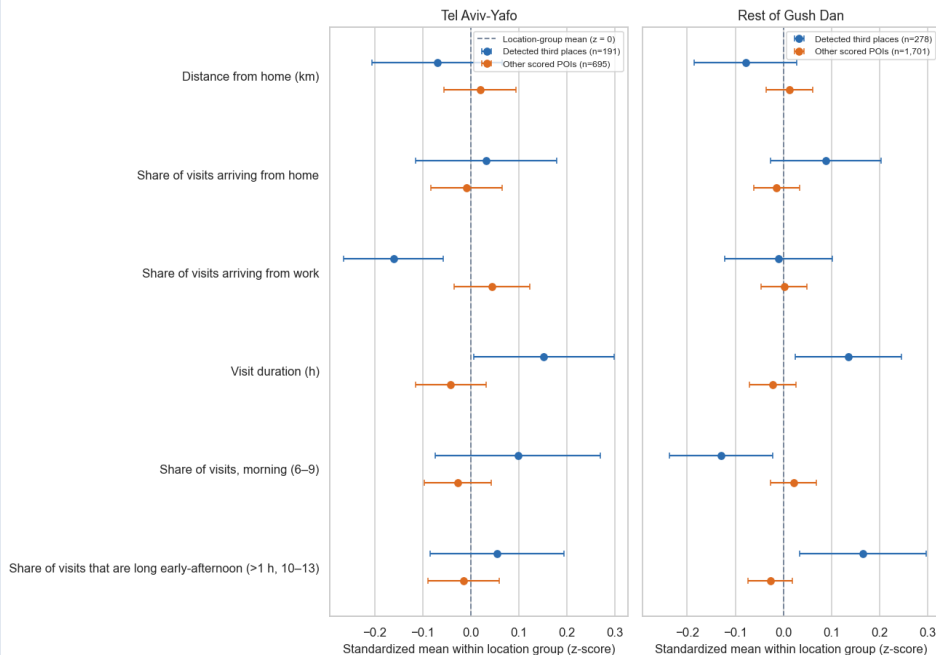
Model shows stability to highly rank ground truth



Part 2 results. How do visits differ between third places and the rest?

STANDARDIZED BEHAVIOR DIFFERENCES

Detected third places vs other scored POIs, Tel Aviv-Yafo vs rest of Gush Dan



SIGNIFICANT DEVIATIONS

Longer visits $+ 0.15 \sigma$

All Tel Aviv Metropolitan area

Distance from home $- 0.08 \sigma$

Only satellite cities

More long afternoon stays $+ 0.17 \sigma$

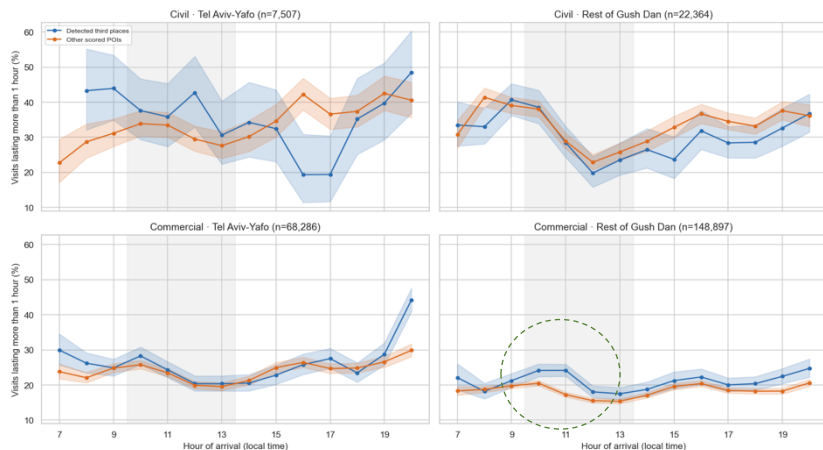
Only satellite cities

Part 2 results · longer visits, closer to home?

Hourly distribution mostly reveal no differences between detected TP and other candidates

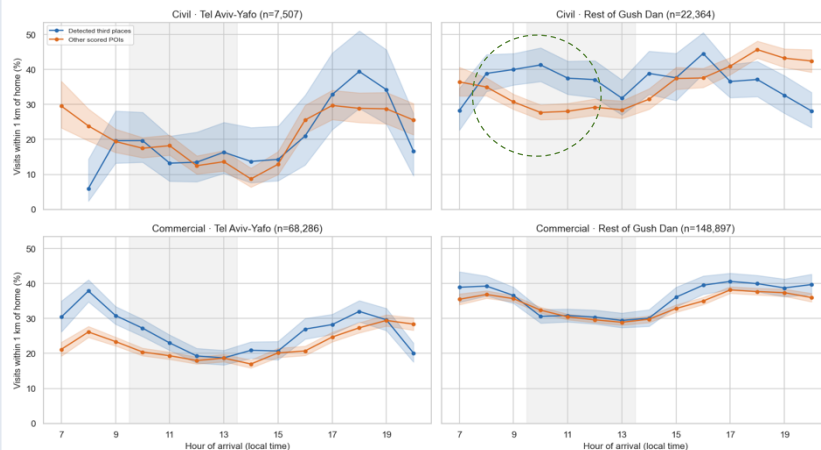
VISITS LASTING > 1 HOUR

Fig. D4 — Share of weekday visits lasting more than one hour, third vs other POIs, civil/commercial × location



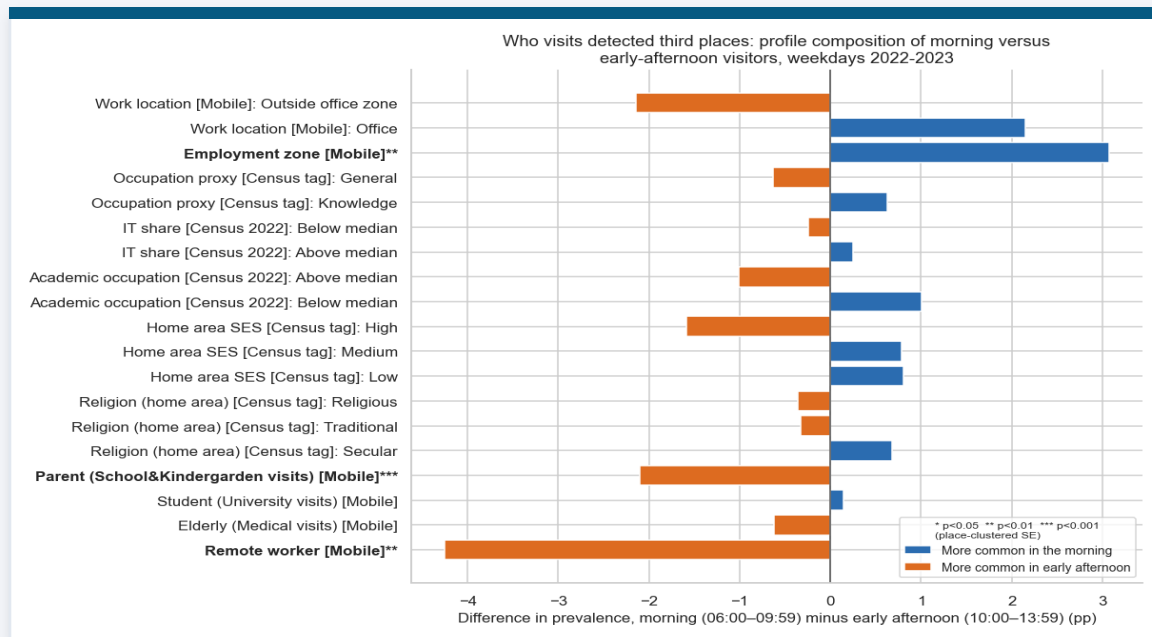
VISITS WITHIN 1 KM FROM HOME

Fig. C4 — Share of weekday visits within 1 km of home, third vs other POIs, civil/commercial × location



Who visits detected third places?

Prevalence of profile groups among morning visits (06:00 – 09:59) minus early-afternoon visits (10:00 – 13:59).



SIGNIFICANT GAPS

Remote workers ** - 4 pp

Strongly under-represented in the morning window — they arrive in the early afternoon.

Parents on school run *** - 2 pp

Strongly under-represented in the morning window — they arrive in the early afternoon.

Employment zone workers ** + 3 pp

Office-anchored users pass through third places on their way to work in the morning.

OLS of group indicator on a morning dummy · SE clustered at place (poi_id) · * p<0.05 ** p<0.01 *** p<0.001

Part 3 results • Do local Third-Places reshape commercial activity Post-COVID?

Term	M1 (Distance)	M2 (WFH_POT)	M3 (Amenity 1km)		M4 (Amenity&WFH_POT)	
	Distance	Capacity	Candidates	TPs	Candidates	TPs
Panel A: Local POI hours / all POI hours						
Post × Dist	-0.004 (0.01)	-0.020 (0.03)	0.029 (0.05)	0.018 (0.03)	0.022 (0.05)	0.009 (0.03)
Post × WFH_POT		-0.0008 (0.001)			-0.0012 (0.005)	-0.0016 (0.005)
Post × Dist × WFH_POT		0.0002 (0.0006)			0.001 (0.002)	0.001 (0.002)
Post × Dose			0.040 (0.03)	0.034 (0.03)	0.043 (0.04)	0.039 (0.04)
Post × Dist × Dose			-0.006 (0.01)	-0.007 (0.01)	-0.007 (0.02)	-0.008 (0.02)
Panel B: Local POI hours / all local hours						
Post × Dist	0.007 (0.00)	0.015 (0.01)	0.080 (0.02)***	0.050 (0.01)***	0.091 (0.02)***	0.047 (0.01)***
Post × WFH_POT		-0.0001 (0.0006)			-0.0025 (0.003)	-0.0035 (0.003)
Post × Dist × WFH_POT		-0.0002 (0.0002)			0.0005 (0.001)	0.001 (0.001)
Post × Dose			0.051 (0.01)***	0.048 (0.01)***	0.065 (0.02)***	0.066 (0.02)***
Post × Dist × Dose			-0.019 (0.01)***	-0.016 (0.01)***	-0.025 (0.01)***	-0.023 (0.01)***
Model Statistics						
Number of Observations	2,420	2,405	2,420	2,420	2,420	2,420
R-squared (Panel A)	0.595	0.595	0.584	0.583	0.584	0.583
R-squared (Panel B)	0.587	0.587	0.699	0.701	0.701	0.703
Time Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Statistical Area Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes
<i>Notes: Evaluated at Dist = 0, where Dist is calculated as ln(Distance + 1). Format: Coef (SE). * p < 0.10, ** p < 0.05, *** p < 0.01.</i>						

A

Regional POIs Substitution —
NOT CONFIRMED

B

The shift of local time from home to
amenities? — CONFIRMED

Part 3 results · Do local Third-Places reshape commercial activity Post-COVID?

Term	M1 (Distance)	M2 (WFH_POT)	M3 (Amenity 1km)		M4 (Amenity&WFH_POT)	
	Distance	Capacity	Candidates	TPs	Candidates	TPs
Panel C: Local candidate hours / all candidate hours						
Post × Dist	—	—	0.205 (0.06)***	0.104 (0.03)***	0.159 (0.06)**	0.093 (0.04)**
Post × WFH_POT					0.008 (0.006)	0.006 (0.007)
Post × Dist × WFH_POT		—			-0.001 (0.003)	-0.0002 (0.003)
Post × Dose			0.139 (0.04)***	0.131 (0.03)***	0.088 (0.05)*	0.096 (0.05)**
Post × Dist × Dose			-0.058 (0.02)***	-0.060 (0.02)***	-0.037 (0.02)*	-0.045 (0.02)**
Panel D: POI hours / all hours						
Post × Dist	-0.004 (0.00)	0.027 (0.01)**	0.068 (0.02)***	0.039 (0.01)***	0.082 (0.02)***	0.037 (0.01)***
Post × WFH_POT		0.0008 (0.0005)			-0.003 (0.002)	-0.004 (0.002)*
Post × Dist × WFH_POT		-0.0006 (0.0002)***			0.0006 (0.0008)	0.001 (0.0008)
Post × Dose			0.050 (0.01)***	0.047 (0.01)***	0.068 (0.01)***	0.067 (0.01)***
Post × Dist × Dose			-0.017 (0.00)***	-0.016 (0.00)***	-0.024 (0.01)***	-0.025 (0.01)***
Panel E: Radius of gyration, POIs (meters)						
Post × Dist	-453.4 (62.2)***	-620.8 (205.3)***	-838.3 (257.8)***	-708.2 (141.0)***	-635.2 (276.1)**	-618.3 (148.8)***
Post × WFH_POT		-12.6 (7.1)*			-23.3 (30.1)	-11.7 (31.6)
Post × Dist × WFH_POT		1.1 (3.4)			-2.1 (13.4)	-6.8 (14.1)
Post × Dose			-255.9 (152.3)*	-282.6 (132.6)**	-87.3 (240.7)	-189.0 (220.2)
Post × Dist × Dose			98.1 (68.5)	108.7 (60.5)*	26.3 (101.7)	63.4 (93.8)
Model Statistics						
Number of Observations	2,420	2,405	2,420	2,420	2,420	2,420
R-squared (Panel C)	—	—	0.546	0.547	0.548	0.549
R-squared (Panel D)	0.641	0.648	0.778	0.779	0.780	0.782
R-squared (Panel E)	0.770	0.775	0.730	0.730	0.733	0.733
Time Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Statistical Area Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Evaluated at Dist = 0, where Dist is calculated as $\ln(\text{Distance} + 1)$.

Format: Coef (SE). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Panel C N = 2,402.

C

The preference for local TP candidates venues —
CONFIRMED

D

The overall affinity for out-of-home activities —
CONFIRMED

E

the commercial travel radius—
PARTIALLY CONFIRMED

Expected outcomes of the study

1

Methodological bridge

First application of Word2Vec embeddings to physical mobility for detecting functional substitution of places.

2

Beyond coworking

Identifies overlooked third-place categories — libraries, cultural centers, museums — not coworking.

3

Local TP matter, but they don't lead to regional substitution

Regression shows local density outperforms WFH capacity in explaining post-COVID relocation of activity in specific POIs categories. However we didn't find any statistically significant impact on total POI consumption

4

Micro-planning input

Third-place map informs where to invest in pedestrian access and Wi-Fi for a true 15-minute city.

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Thank you

Questions & discussion

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APPENDIX



Step 2 · Find embeddings close to 'WORK'

We run 4 Word2Vec model with 2 dictionaries for Pre-Covid and Post-Covid features

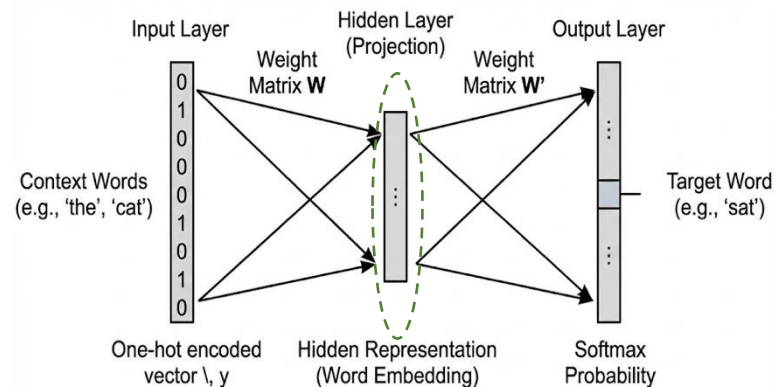
PLACES – WORDS, TRAJECTORIES-SENTENCES

	Home /Work	POI	Unknown location
MODEL I Aggregated locations	'Home'/'Work	'POI'+[POI category]	'GH' + [Day Period]
MODEL II Individual locations	'Home'/'Work' + [Statistical Area ID]	'POI' + [Unique ID] + [Day Period*]	[Geohash Index] + [Day Period]

	TOKENS	FREQUENCY PRE-COVID	FREQUENCY POST_COVID
MODEL I	42	25972	134574
MODEL II	57503		99.8

* Day Period: 'MORNING' (6–9), 'LATE_MORNING' (9–12), 'AFTERNOON' (12–17), 'EVENING' (17–23), 'NIGHT' (23–6)

COS SIM WITH 'WORK' EMBEDDING – A MODEL FEATURE

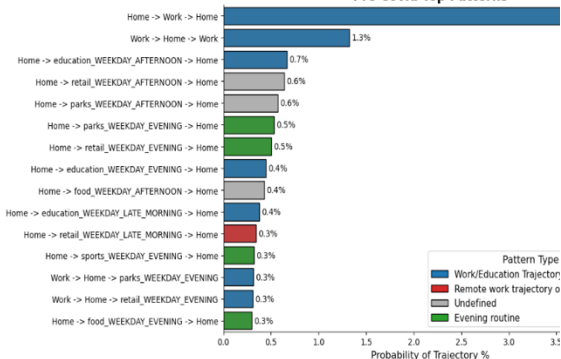


Word2Vec Architecture (Simplified CBOW/Skip-gram)

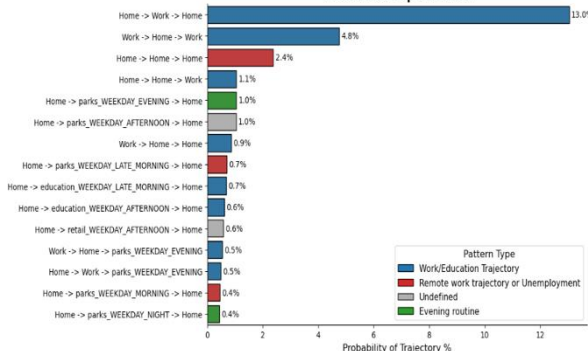
Results • New trajectories occur in 2022-2023 (Model I)

After COVID-19 we start seeing new popular trajectories, indicating remote work behavior

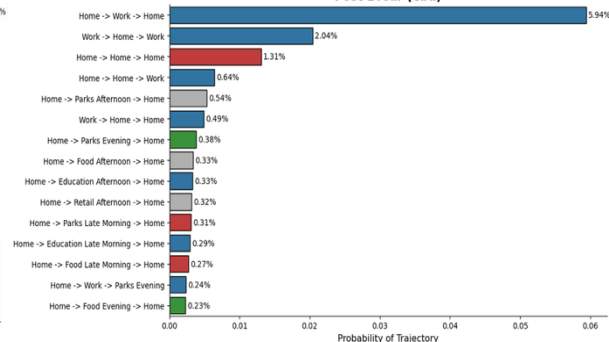
Pre Covid Top Patterns



Post 1Year Top Patterns



Post 1Year (Cbd)



Pre-covid



2022-2023

Closest tokens to 'WORK':

13 / 15